

The VC Brain

System Design Document

Founder-sourcing and investment-intelligence system. Team design reference for a three-person build.

1. Purpose and scope

This document is the single reference the team builds from. It defines the system, the data model, the method for every hard part, the contract between the three layers so three people can work in parallel, the build timeline, and a fully scripted demo path.

The system finds strong founders early, judges each opportunity on evidence rather than network, and produces a decision-ready investment memo fast enough for one investor to act on a small check within a day.

In scope: Sourcing, Screening, Diligence, and Decision.

Out of scope: portfolio monitoring, follow-on, fund operations, and exit. No time is spent on those screens.

2. What we are building

The build rests on three layers that map onto a four-stage pipeline. The required capabilities are fixed by the brief.

- Three layers: Memory, the data foundation. Intelligence, the reasoning and scoring layer. Experience, the investor-facing surface.
- Four pipeline stages: Sourcing, then Screening, then Diligence, then Decision.
- Two sourcing tracks that converge: inbound applications and outbound identification and activation.
- The Founder Score: a persistent per-person score that lives in Memory, persists across every startup a person builds, and never resets.
- The Trust Score: per claim, not per company. Every assertion traces to evidence with a confidence level, is verified against outside sources where possible, and contradictions are flagged before they reach the investor.
- Multi-axis screening: three independent axes, Founder, Market, and Idea versus Market, each with a trend, never averaged into one number.
- Multi-attribute reasoning: a compound natural-language query resolved in one pass, not as separate manual filters.
- The cold-start case: an explicit method for a first-time founder with no funding, little code history, and no network. This is the part the brief singles out, and it is our main differentiator.

3. Build priorities

Effort is allocated to match where the challenge places weight and where the market is least served. The order below is the order of protection when time runs short.

- The cold-start method is treated as a first-class feature, not an add-on. A system that only ranks founders who already leave a public trail rebuilds the network-gated status quo the challenge exists to replace.

Data enters through two sourcing tracks and flows left to right through the pipeline. Underneath, the three layers stack: Memory holds and structures everything, Intelligence reasons over Memory, and Experience presents the result. A feedback loop returns funded outcomes and trends to Memory so scoring sharpens over time.



Sourcing. Opportunities enter two ways. Inbound applicants submit a deck and a company name. Outbound scanning finds founders on public sources and invites them to apply. Both tracks feed one funnel.

Screening. Each opportunity is scored on the three axes, each with a trend. A fast first-pass filter removes clearly non-viable ideas before deeper analysis, so effort is spent where it matters.

Diligence. The Validator checks the claims behind the opportunity against outside evidence, records what is verified, what is unverified, and what contradicts, and logs the gaps.

Decision. The system produces the memo with the ranked axes, the per-claim Trust Scores, and a recommendation, with every conclusion traceable to a source.

6. Layer one: Memory

Memory is the data foundation. Nothing is discarded. It ingests raw signals, resolves them to the right founder and company, removes duplicates, tags each item with its source and timestamp, stores structured records and vector embeddings, and holds the persistent Founder Score.

Responsibilities

- Fetch from heterogeneous sources: GitHub, Crunchbase, arXiv, Product Hunt, Hacker News, and uploaded decks.
- Resolve entities so the same person and company are recognized across sources.
- Deduplicate, enrich, timestamp, and tag every item by source.
- Store structured records in Postgres and text embeddings in pgvector for retrieval.
- Hold the Founder Score as a versioned time series, so the trend is available, not only the latest value.

Ingestion pipeline

Fetch, then normalize to a common signal record, then resolve the entity, then deduplicate, then enrich, then tag, then store and embed. Each signal keeps a link back to its raw source so evidence can be shown later.

7. Layer two: Intelligence

Intelligence reasons over Memory. It is honest about confidence and shows the evidence behind every conclusion.

- Thesis Engine: applies the fund-specific lens, meaning sectors, stage, geography, check size, ownership targets, and risk appetite, to filter and score every opportunity.
- Extraction Agent: reads signals and pulls structured claims about the founder and company, each linked to the signals that support it.
- Validator Agent: cross-references each claim against outside sources, marks it verified, unverified, or contradicted, and produces the per-claim Trust Score. This is the self-correction loop.
- Three-Axis Scorer: scores Founder, Market, and Idea versus Market independently, each with a trend, and never averages them.
- Founder Score Engine: maintains the persistent per-person score and updates it when a founder ships a milestone.
- Cold-Start Module: scores a pre-track-record founder from whatever public footprint exists, and returns a confidence band rather than a false point estimate.
- Memo Generator: assembles the required memo sections, cites the evidence per claim, and marks any gap in plain terms instead of guessing.

- Traceability Logger: records the step-level reasoning behind each conclusion so a recommendation can show exactly what justified it.

8. Layer three: Experience

Experience is the investor-facing surface. It is clear and readable, approachable for a non-technical investor, with analytical depth on demand.

- Thesis Config: the investor sets the fund lens once.
- Application intake: the inbound form, a deck and a company name.
- Ranked Dashboard: the list of opportunities filtered through the thesis, each with a momentum trend.
- Opportunity Detail: the three axes shown separately, each with its trend and rationale, and the claims with their Trust Scores and evidence.
- Memo View: the decision-ready memo with a citation behind every conclusion and gaps marked.
- Founder Profile: the Founder Score over time and the founder's history across startups.
- Outbound Queue: identified founders with their scores and an activation action that triggers an application.

9. The hard parts, solved

9.1 The cold-start method

A first-time founder with no funding, little code history, and no network still leaves a public footprint. The method scores that footprint on traits rather than credentials, and is explicit about how little it knows.

- Collect footprint proxies: written work such as a technical blog, long posts, or forum comments, any open-source contribution however small, competition and hackathon results, academic or research output, and any working artifact such as a live demo.
- Score four traits from the proxies: technical depth from code and artifacts, communication from written work, founder-market fit from domain evidence, and velocity from how consistently the person ships.
- Return a confidence band, not a single number, because the evidence is sparse. A wide band is an honest output, not a failure.
- Separate absence of evidence from negative evidence. No public trail is marked as limited evidence, never as a low score. This is the honesty point that stops the system from punishing people simply for being early.

9.2 The Trust Score method

The Trust Score is attached to each claim, not to the company.

- Each claim carries links to the signals that support it and a count of independent corroborating sources.
- The Validator sets a status: verified when an outside source confirms it, unverified when only the applicant asserts it, and contradicted when an outside source disagrees.
- The confidence level rises with independent corroboration and falls with contradiction or a single unverified source.
- Contradictions are surfaced to the investor, never hidden.

9.3 The three-axis scoring method

- Founder axis: track record, technical depth, team completeness, and the persistent Founder Score as one input.
- Market axis: market size, competitor density, and timing, rated bullish, neutral, or bear.
- Idea versus Market axis: whether the idea survives scrutiny as it stands, or whether the team is strong enough to pivot.
- Each axis shows a trend, meaning improving, declining, or stable, computed from the time series in Memory, and each is presented separately with its rationale.

9.4 The Founder Score method

The Founder Score is a persistent per-person score stored in Memory. It follows a founder across different startups and never resets. It is one input into the Founder axis, not a replacement for it. Each shipped milestone updates it, so a founder who builds again starts from a stronger position. It is stored as a versioned time series so the trend is visible.

9.5 Entity resolution and deduplication

The same person and company appear across many sources under slightly different names. Resolution matches on strong identifiers first, such as a GitHub handle, an email, or a company domain, then falls back to fuzzy name matching combined with embedding similarity on the surrounding text. Matched records are merged, and every merged record keeps its source links so evidence is never lost.

10. Data model

The core entities and their key fields. Every record carries a source and a timestamp so it can be shown as evidence.

Entity	Key fields	Purpose
Founder	id, name, identifiers (GitHub, LinkedIn, email), founder_score_current, history	Persistent person identity across startups
FounderScore	founder_id, value, timestamp, reason	Versioned time series of the persistent score
Opportunity	id, company_name, founder_id, source_track (inbound / outbound), status	One venture under evaluation
Signal	id, opportunity_id, source, url, text, timestamp, confidence	A raw piece of evidence from any source
Claim	id, opportunity_id, statement, category, evidence (signal ids), trust_status, confidence	An extracted assertion with its evidence
ScoreSet	opportunity_id, founder_axis, market_axis, idea_axis, each with value and trend	The three-axis screening result
Memo	opportunity_id, sections, citations, gaps	The decision-ready output
ThesisConfig	sectors, stage, geography, check_size, ownership, risk_appetite	The fund-specific lens
SourcingChannel	id, name, yield, quality	For the network-intelligence stretch

11. Interface contracts between layers

These contracts are agreed in the first hour so the three people can build against stable interfaces and mock what is not ready. The exact field names are frozen early and not changed without telling the other two.

Provider	Operation	Input	Output
Memory	get_opportunity_bundle	opportunity_id	founder, signals, claims, founder_score series
Memory	search_signals	query text or filters	ranked signals with source and timestamp
Memory	write_claims / write_scores	claims or score set	confirmation
Memory	get_founder_history	founder_id	all opportunities and founder_score series
Intelligence	score_opportunity	opportunity_id, thesis_config	score set with trends and rationale
Intelligence	validate_claims	opportunity_id	claims with trust status and confidence
Intelligence	generate_memo	opportunity_id	memo with citations and gaps
Intelligence	cold_start_score	founder footprint	trait scores with confidence band
Experience	consumes read APIs	opportunity_id, thesis_config	renders dashboard, detail, memo, profile

12. Technology stack

- Backend: Python with FastAPI, which suits the agents and the data work.
- Database: Postgres with pgvector, on Supabase, for both structured records and embeddings.
- Reasoning: the OpenAI API for extraction, scoring, validation, and memo generation, with embeddings for deduplication and retrieval. Credits are provided at the event.
- Sourcing fetchers: the GitHub REST API, the Crunchbase API, the arXiv API, the Product Hunt API, and the Hacker News Algolia API.
- Document parsing: OpenAI vision for decks and PDFs into structured fields.
- Frontend: Next.js or React with a simple component library, focused on clarity.
- Traceability: step-level reasoning logs stored per conclusion, shown in the memo and detail views.

13. End-to-end sequence flows

Inbound path

- The investor sets the thesis once in Thesis Config.
- An applicant submits a deck and a company name.
- Memory parses the deck, fetches matching public signals, resolves the founder, deduplicates, tags, and stores.
- The first-pass filter drops clearly non-viable ideas.
- Extraction pulls claims. The Validator sets each claim's trust status and confidence and flags contradictions.
- The Three-Axis Scorer scores the opportunity through the thesis lens, each axis with a trend.

- The Memo Generator writes the memo with citations and marked gaps.
- The opportunity appears on the ranked dashboard, ready for the investor.

Outbound path

- The outbound scanner reads public sources and finds a promising founder.
- The founder is scored the same way as an inbound application, using the cold-start method when the track record is thin.
- The strongest matches enter the Outbound Queue.
- Activation sends a direct invitation to apply. This is outreach, not investment.
- An accepted invitation becomes an application and joins the same screening step as inbound, so both tracks feed one funnel.

14. Task division for three people

Each person owns one layer. The layers match the three people cleanly. The interface contracts in section 11 are agreed first so the three can work in parallel and mock what is not ready.

Person and layer	Owns	Key deliverables	Depends on
Person A: Memory	Ingestion, entity resolution, dedup, storage, Founder Score persistence	Fetchers for two live sources plus deck upload, the store, the read APIs in section 11	Nothing upstream. Delivers the contract first so others can mock.
Person B: Intelligence	Extraction, Validator and Trust Score, three-axis scorer, cold-start module, memo generator, traceability	Scoring, validation, cold-start, and memo behind the Intelligence APIs	Memory read APIs. Mocks them until Person A is live.
Person C: Experience	Thesis config, application form, dashboard, opportunity detail, memo view, founder profile, and integration	The full clickable journey and the demo build	Intelligence and Memory read APIs. Mocks with seed data until they are live.

How the three stay unblocked

- Hour zero to one is spent together agreeing the data model and the interface contracts, and loading the seed data so everyone can mock against real-looking records.
- Person C owns integration and the demo, because the Experience layer is where everything comes together.
- Intelligence is the heaviest layer. Once Memory is stable, Person A moves to help Person B with the Validator and the cold-start module.
- The team syncs briefly at the end of each build block in section 15 to confirm the contracts still hold.

15. Hour-by-hour build timeline

- Hour 0 to 1: agree the data model and interface contracts, set the thesis configuration and the memo structure, create the OpenAI and Supabase accounts, and load the seed data in section 17.
- Hour 1 to 5: Person A builds Memory ingestion from two live sources plus deck upload, with resolution, dedup, tagging, and the read APIs. Person B builds extraction and the three-axis scorer against mocked

Memory. Person C builds the thesis config, the application form, and the dashboard shell against seed data.

- Hour 5 to 10: Person B builds the Validator and the Trust Score and the cold-start module. Person A finishes the Founder Score persistence and helps Person B. Person C builds the opportunity detail and the founder profile.
- Hour 10 to 14: connect the three layers on real data. Person B builds the memo generator and the traceability log. Person C wires the memo view and the outbound queue.
- Hour 14 to 18: run the full inbound and outbound flows end to end. Fix the seams. Tune the scoring and the cold-start band on the seed founders.
- Hour 18 to 21: polish the three screens that appear in the demo, confirm every conclusion in the memo cites a source, and rehearse the demo path in section 16.
- Hour 21 to 24: freeze the build, seed a clean database for the demo, rehearse twice, and prepare the narrative.

16. The demo path

This is the exact run, scene by scene. Each scene lists the starting state, the action to take, what appears on screen, the words to say, and why the scene matters. The whole run is built on the seed founders in section 17. Follow it in order.

Scene 0. Setup, before you present

State: The app is open on the Thesis Config screen. The database is seeded with the four founders in section 17. A second browser tab is open on the Founder Profile for Aria Vance, ready but not shown yet.

Action: Confirm the thesis is empty so you can fill it live.

Why: A clean start makes the live query feel real, not staged.

Scene 1. The thesis and the ranked list

State: Thesis Config screen.

Action: Type the fund lens into the thesis, then enter the compound query in plain language: a technical founder in Berlin, working on AI infrastructure, with enterprise traction, no prior venture backing, and a top-tier accelerator. Submit.

On screen: The ranked dashboard appears. Ligarch, founded by Aria Vance, sits at the top with a momentum trend arrow pointing up. Two weaker opportunities sit below it.

Say: *I set the fund once. Then I ask for exactly the founder I want, in plain language, in a single pass. The system resolves all six conditions together and ranks what fits.*

Why: This proves multi-attribute reasoning in one pass and the thesis lens, and it opens the strongest opportunity for the next scenes.

Scene 2. The three axes, shown separately

State: Ranked dashboard.

Action: Click Ligarch to open the Opportunity Detail.

On screen: Three separate panels. Founder axis: strong, trend improving. Market axis: bullish. Idea versus Market axis: survives as it stands. Each panel shows its own short rationale. There is no single blended number anywhere.

***Say:** The three axes are scored on their own, each with a trend. I never collapse them into one number, because the disagreement between them is exactly what I need to see as an investor.*

Why: This is the not-averaged requirement, shown plainly, with trend.

Scene 3. Evidence, the Trust Score, and a caught contradiction

State: Opportunity Detail for Ligarch.

Action: Scroll to the claims. Point to a verified claim, then open a second opportunity, PulseGrid by Nadia Ksanti, and point to the contradicted claim.

On screen: For Ligarch, the claim of two design partners in production is marked verified and high confidence, with two independent sources linked. For PulseGrid, the deck claim of ten thousand in monthly revenue is marked contradicted, low confidence, with a note that an outside source shows the company is pre-revenue.

***Say:** Every claim carries its own Trust Score and its evidence. Here the traction claim is verified by two independent sources. And here the system caught a contradiction: the deck claims revenue, the outside record says pre-revenue, so it flags it instead of trusting the deck.*

Why: This is the per-claim Trust Score and the Validator self-correction loop, shown with a real catch. It is the most convincing moment for the reasoning score.

Scene 4. The cold-start founder

State: Opportunity Detail.

Action: Open Mateo Rossi, a first-time founder with no funding and almost no code history.

On screen: The Founder axis shows a promising assessment with a visibly wide confidence band. The evidence panel lists a technical blog, a hackathon win, active forum contributions, and a working demo. A clear label reads limited evidence, not negative evidence.

***Say:** This founder has no funding and barely any code history, so most tools would score him near zero or drop him. We score the public footprint instead: his writing, a hackathon win, a working demo. The band is wide on purpose, because we are honest about how little we know. And we mark this as limited evidence, never as a low score. This is the founder the system exists to find.*

Why: This is the cold-start method, the differentiator, and it directly answers the question the brief singles out. It also shows the confidence band and the honesty about uncertainty.

Scene 5. The Founder Score across two startups

State: Opportunity Detail for Ligarch.

Action: Click Aria Vance's name to open the Founder Profile in the second tab.

On screen: A timeline. Aria's Founder Score sits higher today than it did two years ago. The profile shows a previous startup, Vane, which shipped and wound down, and the current one, Ligarch. The score stepped up when Vane shipped and never reset.

***Say:** The Founder Score follows the person, not the company. Aria built Vane two years ago. It shipped, then wound down, and her score stepped up and stayed up. So when she starts Ligarch, she starts from a stronger position. The system never forgets what someone has built.*

Why: This is the persistent Founder Score, shown across two startups over time, and it is what makes the system compound.

Scene 6. The memo and the citations

State: Founder Profile.

Action: Return to Ligarch and open the Memo View.

On screen: The memo with its required sections: company snapshot, investment hypotheses, SWOT, problem and product, and traction and KPIs. Every conclusion has a small citation that links to the exact signal behind it. One section reads cap table not disclosed, marked as a gap rather than filled in.

Say: *The memo reads like something an investor can act on today. Every line traces to its source, so nothing is taken on faith. And where we do not have data, we say so plainly, here on the cap table, rather than guess. A memo that marks its own gaps is the one you can trust.*

Why: This is the decision-ready output, the agentic traceability, and the honest handling of gaps, all of which the memo requirements call for.

Scene 7. Outbound, closing the loop

State: Memo View.

Action: Open the Outbound Queue and select Mateo Rossi, then click Activate.

On screen: Mateo appears in the queue with his cold-start assessment. Activate composes a direct invitation to apply and marks his status as invited.

Say: *The system does not wait for founders to find us. It scans public sources, scores people like Mateo the same way as an applicant, and reaches out to invite the strongest to apply. That invitation becomes an application, and it enters the exact same pipeline. Both tracks feed one funnel.*

Why: This closes the loop and shows the outbound track and convergence, the part of the brief with the least commercial competition.

Closing line

Say: *So the same system finds a founder before anyone is looking, judges the opportunity on evidence rather than network, catches its own mistakes, is honest about what it does not know, and hands the investor a decision they can act on within a day. That is the system we wish had found us.*

17. Pre-seeded demo data

Load these four records before the demo. They are fictional and are the backbone of the run in section 16.

Founder and company	Role in the demo	Key seeded details
Aria Vance, Ligarch	The strong match and the Founder Score story	Technical founder in Berlin, AI infrastructure, two design partners in production verified by two sources, previous startup Vane shipped and wound down, Founder Score risen from 61 to 82
Nadia Ksanti, PulseGrid	The caught contradiction	Deck claims ten thousand in monthly revenue, outside record shows pre-revenue, claim marked contradicted and low confidence

Founder and company	Role in the demo	Key seeded details
Mateo Rossi, no company yet	The cold-start founder and outbound activation	First-time founder, no funding, minimal code history, strong public footprint: technical blog, hackathon win, forum activity, working demo, wide confidence band
A weaker opportunity	Fills the ranked list below Ligarch	Off-thesis or thin evidence, so it ranks lower and shows the ranking works

18. Risks and mitigations

- Competing on breadth. We do not try to out-database the incumbents. We compete on the cold-start method and per-claim traceability.
- Shallow reasoning over shallow data. We build Memory deep first, then a thin but transparent Intelligence layer on top.
- Hallucinated claims. The Validator cross-references every claim against outside sources and marks gaps rather than filling them.
- Averaging the axes. We keep the three axes separate. Collapsing them hides the disagreement an investor needs.
- Live demo fragility. The database is frozen and seeded before the demo, and the run in section 16 is rehearsed twice on that exact data.
- Over-collecting inputs. The application stays a deck and a company name. Fields are added only when a decision needs them.

19. Definition of done

- The thesis lens is configurable and a compound natural-language query returns a ranked list in one pass.
- Memory ingests from at least two live sources plus deck upload, with resolution, dedup, and source tagging.
- Each opportunity shows three separate axes, each with a trend, never averaged.
- Every claim shows a Trust Score with its evidence, and at least one contradiction is caught and flagged.
- The cold-start method scores a founder with no funding or code history from a public footprint, with a confidence band and the limited-evidence label.
- The Founder Score persists across two startups for one founder and shows the trend.
- The memo produces the required sections, cites the source behind every conclusion, and marks gaps.
- The outbound queue identifies a founder and activates an invitation that would enter the same pipeline.